
QUANTUM MOLECULAR HAMILTONIAN

From Lab-Frame Cartesian to Total-CoM
Coordinates

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Contents

1	Quantum Hamiltonian in Cartesian coordinates	3
2	Total-CoM coordinates and the transformed wavefunction	3
2.1	New coordinates	3
2.2	The redundancy and how the wavefunction handles it	4
3	Chain rule: first derivatives	4
3.1	Jacobian of the coordinate change	4
3.2	Chain rule, written out once in components	4
3.3	Cross-check: the total momentum	5
4	Second derivatives and the transformed Hamiltonian	5
4.1	Laplacians	5
4.2	The potential	6
4.3	Transformed Hamiltonian	6
4.4	Consistency: $\hat{H}$ preserves the auxiliary condition	6
4.5	Separation of the CoM motion	7
4.6	Remark: why no Podolsky machinery is needed here	7

1 Quantum Hamiltonian in Cartesian coordinates

As in the classical treatment, we have N_{nuc} nuclei labeled by j with masses M_j , charges $Z_j e$, and positions $\mathbf{R}_j$; and N_{elec} electrons labeled by i with mass m_e , charge $-e$, and positions $\mathbf{r}_i$. All vectors live in a space-fixed (SF) lab frame. The state is a wavefunction $\Psi(\mathbf{R}_1, \dots, \mathbf{R}_{N_{\text{nuc}}}, \mathbf{r}_1, \dots, \mathbf{r}_{N_{\text{elec}}})$, and canonical quantization in Cartesian coordinates is the substitution $\mathbf{P}_j \rightarrow \hat{\mathbf{P}}_j = -i\hbar \nabla_{\mathbf{R}_j}$, $\mathbf{p}_i \rightarrow \hat{\mathbf{p}}_i = -i\hbar \nabla_{\mathbf{r}_i}$, giving

$$\hat{H} = \underbrace{-\sum_{j=1}^{N_{\text{nuc}}} \frac{\hbar^2}{2M_j} \nabla_{\mathbf{R}_j}^2}_{\hat{T}_N} - \underbrace{\sum_{i=1}^{N_{\text{elec}}} \frac{\hbar^2}{2m_e} \nabla_{\mathbf{r}_i}^2}_{\hat{T}_e} + V, \quad (1.1)$$

where $\nabla_{\mathbf{R}_j}^2 = \sum_{\alpha=x,y,z} \partial^2 / \partial R_{j\alpha}^2$ and, writing $R_{ij} = |\mathbf{R}_i - \mathbf{R}_j|$ etc.,

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{2} \sum_{i \neq j} \frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j} \frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{r_{ij}} \right). \quad (1.2)$$

The canonical commutators are $[\hat{R}_{j\alpha}, \hat{P}_{k\beta}] = i\hbar \delta_{jk} \delta_{\alpha\beta}$, $[\hat{r}_{i\alpha}, \hat{p}_{l\beta}] = i\hbar \delta_{il} \delta_{\alpha\beta}$, and the inner product is the flat Cartesian one,

$$\langle \Phi | \Psi \rangle = \int \Phi^* \Psi \prod_j d^3 R_j \prod_i d^3 r_i, \quad (1.3)$$

with respect to which each $-i\hbar \partial$ is Hermitian by integration by parts (wavefunctions vanishing at infinity). No ordering ambiguity arises in (1.1): the masses are constants, so the kinetic operators are unambiguous.

The task of this note is purely computational: re-express the *derivatives* in (1.1) in total-CoM coordinates using the chain rule, and watch the CoM motion separate.

2 Total-CoM coordinates and the transformed wavefunction

2.1 New coordinates

Exactly as in the classical treatment, define the *total* (nuclei + electrons) CoM and refer every particle to it:

$$\mathbf{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}} \left(\sum_j M_j \mathbf{R}_j + m_e \sum_i \mathbf{r}_i \right), \quad M_{\text{tot}} = \underbrace{\sum_j M_j + N_{\text{elec}} m_e}_{\equiv M_N}, \quad (2.1)$$

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{cm}}, \quad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{cm}}. \quad (2.2)$$

Multiplying each equation in (2.2) by its mass and summing, the definition (2.1) forces the linear relation

$$\sum_j M_j \mathbf{R}'_j + m_e \sum_i \mathbf{r}'_i = 0. \quad (2.3)$$

2.2 The redundancy and how the wavefunction handles it

Count coordinates. The old set has $3(N_{\text{nuc}} + N_{\text{elec}})$ components; the new set $(\mathbf{R}_{\text{cm}}, \{\mathbf{R}'_j\}, \{\mathbf{r}'_i\})$ has $3(N_{\text{nuc}} + N_{\text{elec}} + 1)$, three too many, and (2.3) is precisely the compensating relation. We express the wavefunction in the new variables,

$$\Psi(\mathbf{R}_j, \mathbf{r}_i) = \psi(\mathbf{R}_{\text{cm}}, \{\mathbf{R}'_j\}, \{\mathbf{r}'_i\}), \quad (2.4)$$

but because the new arguments are constrained by (2.3), equation (2.4) does not define ψ *off* the constraint set: ψ is free there, and every choice represents the same physical Ψ . Before we can differentiate ψ with respect to its arguments *as if they were independent* — which is what the chain rule requires — we must complete its definition. Introduce the total internal-gradient operator

$$\mathbf{D} \equiv \sum_k \nabla_{\mathbf{R}'_k} + \sum_l \nabla_{\mathbf{r}'_l}, \quad (2.5)$$

which generates an *equal shift* of all relative vectors, $\mathbf{R}'_k \rightarrow \mathbf{R}'_k + \boldsymbol{\varepsilon}$, $\mathbf{r}'_l \rightarrow \mathbf{r}'_l + \boldsymbol{\varepsilon}$, and complete the definition of ψ by the auxiliary condition

$$\boxed{\mathbf{D} \psi = 0}. \quad (2.6)$$

This is consistent: under the equal shift the constraint function $\sum_j M_j \mathbf{R}'_j + m_e \sum_i \mathbf{r}'_i$ changes by $M_{\text{tot}} \boldsymbol{\varepsilon} \neq 0$, so the shift moves *transversally off* the constraint set — condition (2.6) only prescribes how ψ continues off the set and places no restriction on its physical values on it. Equation (2.6) is the quantum image of the classical dual constraint $\sum_k \mathbf{P}'_k + \sum_l \mathbf{p}'_l = 0$: acting with $-i\hbar$, it reads $(\sum_k \hat{\mathbf{P}}'_k + \sum_l \hat{\mathbf{p}}'_l) \psi = 0$.

3 Chain rule: first derivatives

3.1 Jacobian of the coordinate change

Every new coordinate in (2.1)–(2.2) is a *linear* function of the old ones, so all partial derivatives of new with respect to old are constants, diagonal in the Cartesian index. Differentiating (2.1)–(2.2) with respect to a nuclear position,

$$\frac{\partial \mathbf{R}_{\text{cm}}}{\partial \mathbf{R}_j} = \frac{M_j}{M_{\text{tot}}} \mathbb{1}, \quad \frac{\partial \mathbf{R}'_k}{\partial \mathbf{R}_j} = \left(\delta_{jk} - \frac{M_j}{M_{\text{tot}}} \right) \mathbb{1}, \quad \frac{\partial \mathbf{r}'_l}{\partial \mathbf{R}_j} = -\frac{M_j}{M_{\text{tot}}} \mathbb{1}, \quad (3.1a)$$

and with respect to an electron position,

$$\frac{\partial \mathbf{R}_{\text{cm}}}{\partial \mathbf{r}_i} = \frac{m_e}{M_{\text{tot}}} \mathbb{1}, \quad \frac{\partial \mathbf{R}'_k}{\partial \mathbf{r}_i} = -\frac{m_e}{M_{\text{tot}}} \mathbb{1}, \quad \frac{\partial \mathbf{r}'_l}{\partial \mathbf{r}_i} = \left(\delta_{il} - \frac{m_e}{M_{\text{tot}}} \right) \mathbb{1}. \quad (3.1b)$$

3.2 Chain rule, written out once in components

For one Cartesian component α of one nuclear position, the chain rule applied to (2.4) reads

$$\frac{\partial \Psi}{\partial R_{j\alpha}} = \sum_{\beta} \frac{\partial R_{\text{cm},\beta}}{\partial R_{j\alpha}} \frac{\partial \psi}{\partial R_{\text{cm},\beta}} + \sum_k \sum_{\beta} \frac{\partial R'_{k\beta}}{\partial R_{j\alpha}} \frac{\partial \psi}{\partial R'_{k\beta}} + \sum_l \sum_{\beta} \frac{\partial r'_{l\beta}}{\partial R_{j\alpha}} \frac{\partial \psi}{\partial r'_{l\beta}}. \quad (3.2)$$

All the Jacobian blocks (3.1) are proportional to $\mathbb{1}$, i.e. to $\delta_{\alpha\beta}$, so the β -sums collapse and the three vector components assemble into

$$\nabla_{\mathbf{R}_j} \Psi = \left[\frac{M_j}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} + \nabla_{\mathbf{R}'_j} - \frac{M_j}{M_{\text{tot}}} \mathbf{D} \right] \psi, \quad (3.3)$$

where the $-(M_j/M_{\text{tot}})\mathbf{D}$ collects the $-M_j/M_{\text{tot}}$ entries that appear in *every* relative coordinate in (3.1a). The same steps for an electron position give

$$\nabla_{\mathbf{r}_i} \Psi = \left[\frac{m_e}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} + \nabla_{\mathbf{r}'_i} - \frac{m_e}{M_{\text{tot}}} \mathbf{D} \right] \psi. \quad (3.4)$$

Now use the auxiliary condition (2.6). All the operators in (3.3)–(3.4) have *constant* coefficients, so $\mathbf{D}$ commutes with each of them; hence if $\mathbf{D}\psi = 0$, then $\mathbf{D}$ annihilates every constant-coefficient derivative of ψ as well, and the $\mathbf{D}$ -terms may be dropped *inside any subsequent derivative*:

$$\boxed{\nabla_{\mathbf{R}_j} \Psi = \left[\frac{M_j}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} + \nabla_{\mathbf{R}'_j} \right] \psi, \quad \nabla_{\mathbf{r}_i} \Psi = \left[\frac{m_e}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} + \nabla_{\mathbf{r}'_i} \right] \psi.} \quad (3.5)$$

Multiplying (3.5) by $-i\hbar$ shows it is precisely the operator image of the classical momentum relation: $\hat{\mathbf{P}}_j = (M_j/M_{\text{tot}})\hat{\mathbf{P}}_{\text{cm}} + \hat{\mathbf{P}}'_j$ and $\hat{\mathbf{p}}_i = (m_e/M_{\text{tot}})\hat{\mathbf{P}}_{\text{cm}} + \hat{\mathbf{p}}'_i$, with $\hat{\mathbf{P}}_{\text{cm}} = -i\hbar\nabla_{\mathbf{R}_{\text{cm}}}$ etc.

3.3 Cross-check: the total momentum

Sum (3.3) over all nuclei and (3.4) over all electrons *before* imposing (2.6). The CoM coefficients add to $(M_N + N_{\text{elec}}m_e)/M_{\text{tot}} = 1$; the plain relative gradients add to $\mathbf{D}$; and the $\mathbf{D}$ -corrections add to $-[(M_N + N_{\text{elec}}m_e)/M_{\text{tot}}]\mathbf{D} = -\mathbf{D}$. The last two cancel *identically* — no auxiliary condition needed:

$$\left(\sum_j \nabla_{\mathbf{R}_j} + \sum_i \nabla_{\mathbf{r}_i} \right) \Psi = \nabla_{\mathbf{R}_{\text{cm}}} \psi \quad \Longleftrightarrow \quad \sum_j \hat{\mathbf{P}}_j + \sum_i \hat{\mathbf{p}}_i = \hat{\mathbf{P}}_{\text{cm}}. \quad (3.6)$$

The total momentum operator is exactly the CoM gradient, independent of how ψ was extended off the constraint set.

4 Second derivatives and the transformed Hamiltonian

4.1 Laplacians

Relation (3.5) expresses $\nabla_{\mathbf{R}_j}$, acting on any function written in the new coordinates, as a fixed constant-coefficient combination of new-coordinate derivatives. Applying it twice — legitimate precisely because the coefficients are constants, and mixed partial derivatives of smooth functions commute — and expanding the square,

$$\begin{aligned} \nabla_{\mathbf{R}_j}^2 \Psi &= \left[\frac{M_j}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} + \nabla_{\mathbf{R}'_j} \right] \cdot \left[\frac{M_j}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} + \nabla_{\mathbf{R}'_j} \right] \psi \\ &= \left[\frac{M_j^2}{M_{\text{tot}}^2} \nabla_{\mathbf{R}_{\text{cm}}}^2 + \frac{2M_j}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} \cdot \nabla_{\mathbf{R}'_j} + \nabla_{\mathbf{R}'_j}^2 \right] \psi, \end{aligned} \quad (4.1)$$

and identically for the electrons with $M_j \rightarrow m_e$, $\mathbf{R}'_j \rightarrow \mathbf{r}'_i$. Assemble the nuclear kinetic operator, using $\sum_j M_j = M_N$ in the first term:

$$\hat{T}_N \Psi = - \sum_j \frac{\hbar^2}{2M_j} \nabla_{\mathbf{R}_j}^2 \Psi = \left[-\frac{M_N}{M_{\text{tot}}} \frac{\hbar^2}{2M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}}^2 - \frac{\hbar^2}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} \cdot \sum_j \nabla_{\mathbf{R}'_j} - \sum_j \frac{\hbar^2}{2M_j} \nabla_{\mathbf{R}'_j}^2 \right] \psi, \quad (4.2)$$

$$\hat{T}_e \Psi = - \sum_i \frac{\hbar^2}{2m_e} \nabla_{\mathbf{r}_i}^2 \Psi = \left[-\frac{N_{\text{elec}} m_e}{M_{\text{tot}}} \frac{\hbar^2}{2M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}}^2 - \frac{\hbar^2}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} \cdot \sum_i \nabla_{\mathbf{r}'_i} - \sum_i \frac{\hbar^2}{2m_e} \nabla_{\mathbf{r}'_i}^2 \right] \psi. \quad (4.3)$$

Adding (4.2) and (4.3): the two mass fractions in front of $\nabla_{\mathbf{R}_{\text{cm}}}^2$ sum to $(M_N + N_{\text{elec}} m_e)/M_{\text{tot}} = 1$, and the two cross terms combine into the total internal gradient (2.5),

$$-\frac{\hbar^2}{M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}} \cdot \underbrace{\left(\sum_j \nabla_{\mathbf{R}'_j} + \sum_i \nabla_{\mathbf{r}'_i} \right)}_{=\mathbf{D}} \psi = 0 \quad \text{by the auxiliary condition (2.6),} \quad (4.4)$$

where dropping $\mathbf{D}$ under the extra $\nabla_{\mathbf{R}_{\text{cm}}}$ is again justified because $\mathbf{D}$ commutes with constant-coefficient derivatives. No CoM-internal cross-coupling survives.

4.2 The potential

V is a multiplication operator, so no derivatives act on it during the coordinate change; we only rewrite its arguments. Subtracting (2.2) pairwise, $\mathbf{R}'_j - \mathbf{R}'_k = \mathbf{R}_j - \mathbf{R}_k$, $\mathbf{R}'_j - \mathbf{r}'_i = \mathbf{R}_j - \mathbf{r}_i$, $\mathbf{r}'_i - \mathbf{r}'_l = \mathbf{r}_i - \mathbf{r}_l$: every interparticle separation in (1.2) equals the corresponding primed separation, so V is unchanged in form,

$$V = V(\mathbf{R}'_j, \mathbf{r}'_i). \quad (4.5)$$

4.3 Transformed Hamiltonian

Collecting (4.2)–(4.5), the Schrödinger operator in total-CoM coordinates is

$$\boxed{\hat{H} \psi = \left[-\frac{\hbar^2}{2M_{\text{tot}}} \nabla_{\mathbf{R}_{\text{cm}}}^2 - \sum_j \frac{\hbar^2}{2M_j} \nabla_{\mathbf{R}'_j}^2 - \sum_i \frac{\hbar^2}{2m_e} \nabla_{\mathbf{r}'_i}^2 + V(\mathbf{R}'_j, \mathbf{r}'_i) \right] \psi, \quad \mathbf{D} \psi = 0.} \quad (4.6)$$

Exactly as in the classical result: the CoM kinetic term carries the correct total mass M_{tot} , the internal kinetic energy is *diagonal* — no mass-polarization term — and no cross-coupling appears, for *any* CoM motion. The price is the constraint pair, (2.3) on the coordinates and (2.6) on the wavefunction.

4.4 Consistency: $\hat{H}$ preserves the auxiliary condition

Condition (2.6) is only meaningful if $\hat{H}$ maps the space $\{\psi : \mathbf{D} \psi = 0\}$ into itself, i.e. if $[\mathbf{D}, \hat{H}] = 0$. The kinetic terms in (4.6) are constant-coefficient derivatives, which commute with $\mathbf{D}$. For the potential, $\mathbf{D}$ is a first-order operator, so $[\mathbf{D}, V] \psi = (\mathbf{D} V) \psi$, and

$$\mathbf{n} \cdot \mathbf{D} V = \mathbf{n} \cdot \left(\sum_k \nabla_{\mathbf{R}'_k} V + \sum_l \nabla_{\mathbf{r}'_l} V \right) = \frac{d}{d\varepsilon} V(\mathbf{R}'_j + \varepsilon \mathbf{n}, \mathbf{r}'_i + \varepsilon \mathbf{n}) \Big|_{\varepsilon=0} = 0 \quad (4.7)$$

for every direction $\mathbf{n}$, because by (4.5) V depends only on *differences* of the primed vectors, which the equal shift leaves untouched; hence $\mathbf{D}V = 0$. Hence $[\mathbf{D}, \hat{H}] = 0$: a wavefunction satisfying (2.6) keeps satisfying it under the dynamics.

4.5 Separation of the CoM motion

$\hat{H}$ in (4.6) contains $\mathbf{R}_{\text{cm}}$ only through $\nabla_{\mathbf{R}_{\text{cm}}}^2$, so it commutes with $\hat{\mathbf{P}}_{\text{cm}} = -i\hbar\nabla_{\mathbf{R}_{\text{cm}}}$ and the ansatz

$$\psi = e^{i\mathbf{K}\cdot\mathbf{R}_{\text{cm}}} \chi(\{\mathbf{R}'_j\}, \{\mathbf{r}'_i\}) \quad (4.8)$$

separates exactly: every $\nabla_{\mathbf{R}_{\text{cm}}}$ pulls down $i\mathbf{K}$ and touches nothing else (note $\mathbf{D}$ does not act on $\mathbf{R}_{\text{cm}}$, so $\mathbf{D}\psi = 0 \Leftrightarrow \mathbf{D}\chi = 0$). The eigenvalue problem $\hat{H}\psi = E\psi$ becomes

$$\boxed{\hat{H}_{\text{int}} \chi = \left[-\sum_j \frac{\hbar^2}{2M_j} \nabla_{\mathbf{R}'_j}^2 - \sum_i \frac{\hbar^2}{2m_e} \nabla_{\mathbf{r}'_i}^2 + V(\mathbf{R}'_j, \mathbf{r}'_i) \right] \chi = \left(E - \frac{\hbar^2 K^2}{2M_{\text{tot}}} \right) \chi, \quad \mathbf{D}\chi = 0,} \quad (4.9)$$

a free particle of mass M_{tot} carrying the CoM, times an internal problem whose kinetic energy is diagonal.

4.6 Remark: why no Podolsky machinery is needed here

The transformation (2.1)–(2.2) is *linear with constant coefficients*. Choosing any independent subset of the new coordinates (say, dropping one relative vector via (2.3)), the old-to-new map is an invertible linear map with a *constant* Jacobian determinant, so the Cartesian volume element (1.3) goes over to the flat new-coordinate volume element times an irrelevant constant. The weight function in the inner product stays constant, $-i\hbar\partial$ remains Hermitian by the same integration by parts as in Section 1, and the naive substitution rule survives the coordinate change untouched — which is why plain chain-rule calculus sufficed here. This is the essential contrast with *curvilinear* coordinates (Euler angles, normal coordinates), where the Jacobian depends on the coordinates, the volume element acquires a nontrivial weight $\sqrt{g}$, and the kinetic operator must instead be built from the Laplace–Beltrami (Podolsky) construction, as carried out in the main text.